

ed image or camera sensors with newer technologies like RGB-IR sensors for better low-light sensitivity and faster processors like ARM Cortex-A series processors to handle more computationally-intensive vision processing.

Newcomers like STM32F4x9 with HW graphic acceleration (Chrom-ART accelerator) and MIPI support are good for applications with displays. Embedded hardware with PDM-to-PCM conversion (DF-SDM on STM32L4), together with low-power batch-acquisition mode, allows voice-recognition feature to run efficiently on STM32 platform.

Driving ultra low power

With significant integration and lower cost requirements, MCUs have been going down process nodes more aggressively than in the past.

Navada explains that most MCUs announced in the last 18 months have brought power consumption down by going to deeper sub-micron processes and adding features such as power shut-off. Wireless connectivity and MCUs are now showing up either in a single package or on monolithic silicon as MCUs and connectivity are going hand in hand. Finally, power management features such as buck converters are getting integrated into MCUs to allow better power efficiency.

As mentioned above, the fact that dynamic power and area reduce at advanced geometries and aggressive integration requirements, lower process node such as 55nm and below are becoming common place. "Requirement of integrated flash puts a brake on going down all the way to the most aggressive process geometries used by microprocessors. Aggressive power shut off, multi-voltage and low-voltage designs are some of the architectural changes seen in reducing power consumption," adds Navada.

Martins further explains that

What is the difference between CPUs and MCUs?

Versatility of MCUs is being proven yet again as MCUs find their way into new applications that are driving semiconductor growth. These play a major role not only in terms of speed and performance, but also in terms of intelligence at a higher level. These are the primary broad advances in the new MCUs. Unlike CPUs, which advance technology node, MCUs with their peripherals and multi-functional capabilities have to address a wide range of applications used in different industries. For example, in automotive, MCUs have to be reliable, as these are high-performing. AURIX MCU for automotive has ten cores, four of which are core-checkers for ensuring that six multi-cores are properly working at all times in the harsh automotive environment. With advances in technology and small node size, cost difference between 32-bit and 8-bit or 16-bit is becoming less significant. It is foreseeable that 32-bit architecture is becoming the mainstay architecture for MCUs.

—Vishal Malhotra, national sales head - automotive, Infineon Technologies, India, and Andy Wong, senior regional marketing manager for Industrial Microcontroller (IMC), PMM division, Infineon Singapore APAC

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process technology improvements (going towards smaller technology lithography) tend to further lower internal core voltage to minimise dynamic power consumption, but not without control over leakage current responsible for consumption in low-power/standby mode. He adds, "Power consumption cannot be judged only from datasheet values but rather depends on real application profile results. There are many embedded MCU features that contribute to optimise power consumption with batch-acquisition mode, ART accelerator and DMIPS (to execute the same code in less time, which leads to less power consumption), highly-flexible clock speed and source selection, along with scalable internal voltage."

The future

As we already know, the line between MCUs and SoCs is getting blurred. Looking at the advances that have happened, blurred seems to be an understatement. Cores are now available from firms like Imagination Technologies, which allow MCU or embedded controller developers to build MCUs that can go so far as to execute full-blown OSes like Linux.

When can you start using such chips in your project? For starters, Microchip has already announced on their website that the CPU at the heart of their PIC32MZ family has been upgraded to a MIPS M5150—a Warrior M-class processor core that comes with full hardware virtualisation. **EFY**